CSNS RCS IPM — Electron-mode and ion-mode results

Code: Virtual-IPM 2.3.1. **Reference:** Rehman et al., NIM A **1092** (2026) 171809. **Cage:** 220 × 231 mm, design V = 25 kV so E_y = V/d ≈ 108 kV/m; design B_y = 0.1 T. **Statistics:** 100000 secondaries per run; quoted σ is the RMS of **detected** x. Space charge is the Gaussian bunch E plus Lorentz-boosted bunch B, compared with those fields off. Guiding E and B stay uniform.

This report covers the **round-beam** e-mode replan (Blocks A–D + fine C/D) and the parallel **ion-mode** matrix (Blocks A–D). Elliptical 25×20 / 10×8 mm archives remain in REPORT.md §§1–6 as a closed reference; they are not reused here.

1. Setup

	Injection	Extraction
Proton energy	80 MeV	1.6 GeV
Bunch length σ_t	120 ns	20 ns
Round-beam σ_x=σ_y	25 mm (painted) or 10 mm	10 mm
Bunch population at 100 kW	7.8×10^{12}	same, ∝ P

- **Electron mode:** Voitkiv DDCS, GasType Hydrogen only (Virtual-IPM has no N_2/H_2O electron DDCS). Collection at the top electrode. Tracker: Boris, 8000 steps. RNG 1234.
- **Ion mode:** ZeroMomentum ions at rest masses 2 / 18 / 28 u (H_2^+, H_2O^+, N_2^+ — the paper ToF peaks). Collection at the bottom electrode. Bunch fields off in the generator plus a circular 3-bunch train.
- **SC-off** is run once per geometry at 100 kW and shared across power (no bunch field ⇒ independent of bunch population).

Re-run: ./scripts/run_emode_replan.sh, ./scripts/run_emode_fine_cd.sh, ./scripts/run_imode_replan.sh . Evaluate: python3 scripts/evaluate_emode.py --from-summary, python3 scripts/evaluate_imode.py --from-summary.

2. Takeaways

1. **Electron IPM is recoverable with guiding B.** Expansion oscillates with B (focusing / over-kick of opposite-sign electrons). **300 G** keeps every scanned round-beam family, power (100–500 kW), cage voltage (5–30 kV), and Δ y offset (±10 mm) within ≲ 1%. Painted injection recovers first (~105–155 G); extraction and 10 mm injection need ~190–290 G.
2. **Ion IPM is not recovered by B.** Expansion is flat from 0 to 200 G and only slightly lower at 0.1 T. Cyclotron radii at these fields are metres. **Do not use the e-mode B knob on ions.**
3. **Ion distortion scales with power and inversely with beam size.** H_2^+ injection, 10 mm, 0.1 T: +82% at 100 kW → +422% at 500 kW. At σ=3 mm / 500 kW the obtained width saturates on the cage (~56 mm, +1700%+) for all three species.
4. **Cage voltage has opposite roles.** In e-mode at B=0, 100 kW **crosses zero at 13 kV** (expansion → compression); 500 kW never flips. In ion-mode expansion stays **positive** at every V; higher V shortens ToF and **reduces** the kick (H_2^+ inj. 10 mm @ 0.1 T: +198% at 5 kV → +70% at 30 kV).
5. **Offset:** Δ x is translation-invariant in both modes. Δ y (toward/away from the detector) changes expansion by a few percent and does not undo the 300 G e-mode recovery.
6. **Practical implication:** tune B and V for a faithful **electron** profile; treat **ion-mode sizes as space-charge biased** unless a correction is applied.

3. Electron mode

Status: complete. Replan A–D 2502 / 2502; fine C/D 3270 / 3270 new (5772 particle CSVs including A–D). All blocks use **round beams** $\sigma_y=\sigma_x$.

Block	What is scanned	Beams	σ	B	Power
A	Guiding field	inj. 25 mm; ext. 10 mm; inj. 10 mm	round	0–300 G / 5 G	100–500 kW
B	Beam size	inj.; ext.	3–20 mm / 1 mm	0, 100, 200, 300 G, 0.1 T	100–500 kW
C	Cage voltage	inj. 10 mm	round	0–300 G / 25 G + 0.1 T	100, 500 kW
Fine C	Voltage × B	inj. 10 mm	round	diagnostic B + 5 G at 10–25 kV	100, 500 kW
D	Beam offset	inj.; ext. 10 mm	round	0, 100, 200, 300 G, 0.1 T	100, 500 kW
Fine D	Δ y	inj.; ext. 10 mm	round	diagnostic B + 5 G at ±5 mm	100, 500 kW

Diagnostic B: 0, 25, 50, 75, 100, 125, 150, 200, 250, 300, 1000 G. Fine C voltages: 5–30 kV / 1 kV. Fine D: Δ y=-10…+10 mm / 1 mm, Δ x=0.

3.1 Block A — B scan 0–300 G

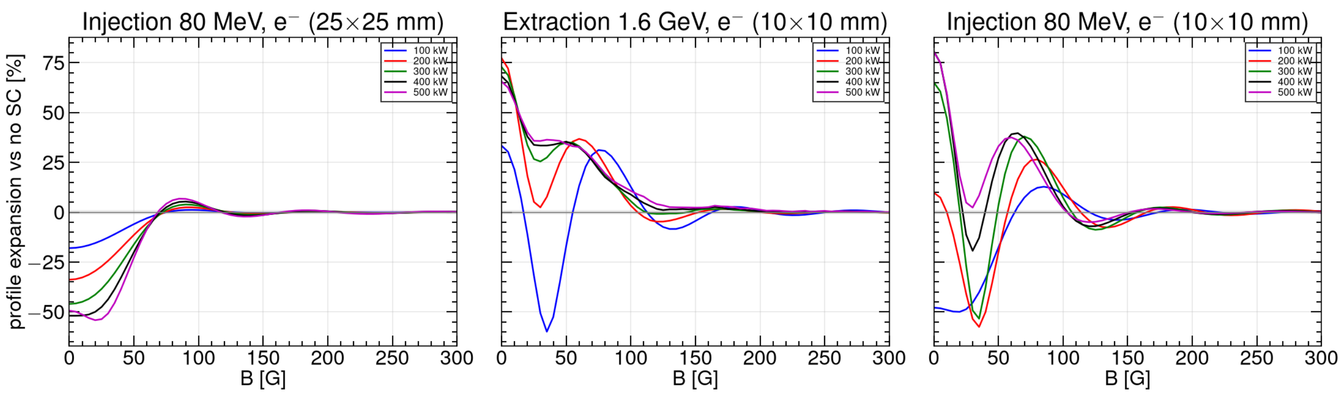

Figure 1. Profile expansion vs no-SC for the three round reference beams, 100–500 kW. The grey band is ±1%.

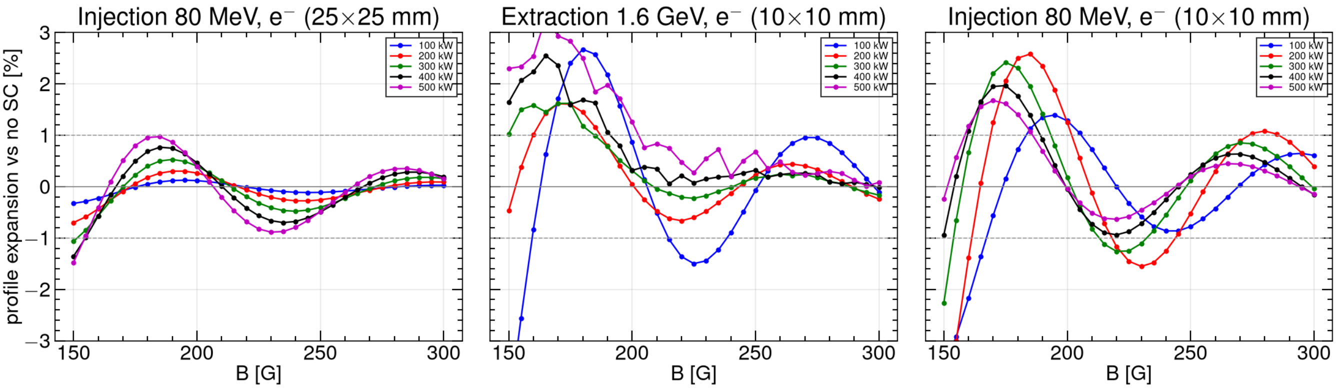

Figure 2. Same scan, 150–300 G, ±3% zoom. Extraction and 10 mm injection still oscillate through ~150–250 G; **300 G** has settled.

Smallest B after which |Δ| stays below 1% for the rest of the grid, and Δ at 300 G:

Family	100 kW	200 kW	300 kW	400 kW	500 kW
Inj. 25×25 mm	105 G (+0.03%)	115 G (+0.09%)	155 G (+0.16%)	155 G (+0.19%)	155 G (+0.15%)
Ext. 10×10 mm	240 G (−0.11%)	190 G (−0.24%)	185 G (−0.17%)	195 G (−0.02%)	205 G (+0.08%)
Inj. 10×10 mm	210 G (+0.61%)	290 G (+0.40%)	235 G (−0.04%)	190 G (−0.16%)	190 G (−0.15%)

Painted injection recovers first. The last curve to settle is 10×10 mm injection at 200 kW (threshold 290 G). **300 G keeps every power ≲ 1%.**

3.2 Block B — size 3–20 mm

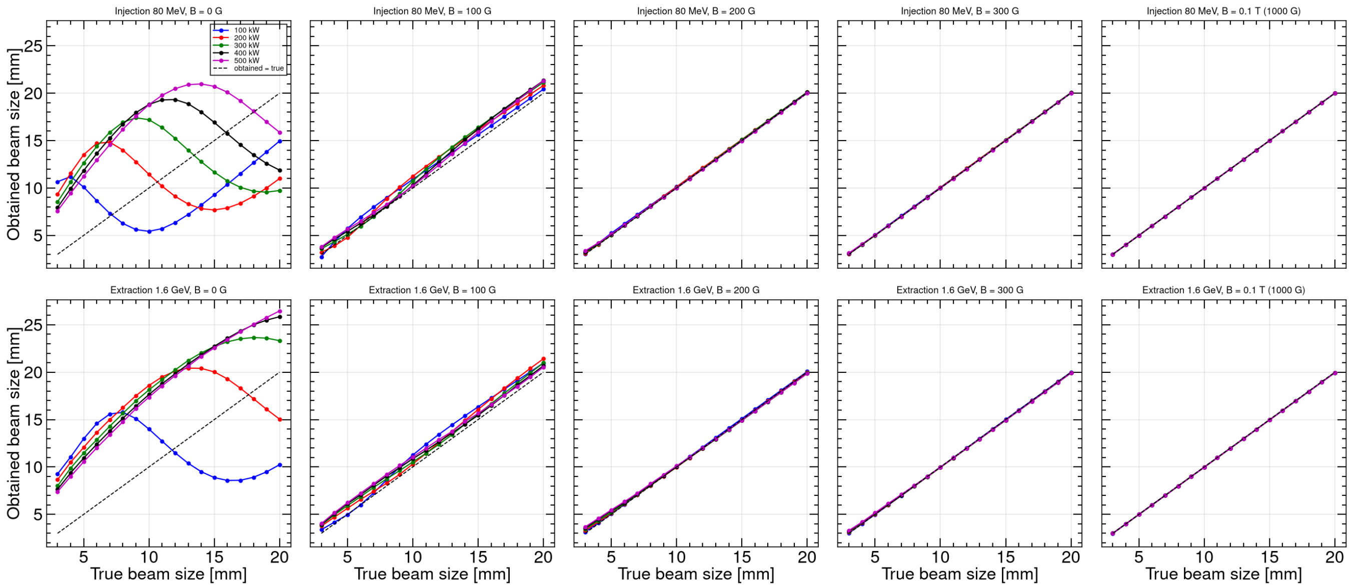

Figure 3. Obtained σ vs true σ at 0, 100, 200, 300 G and 0.1 T. Dashed line: obtained = true.

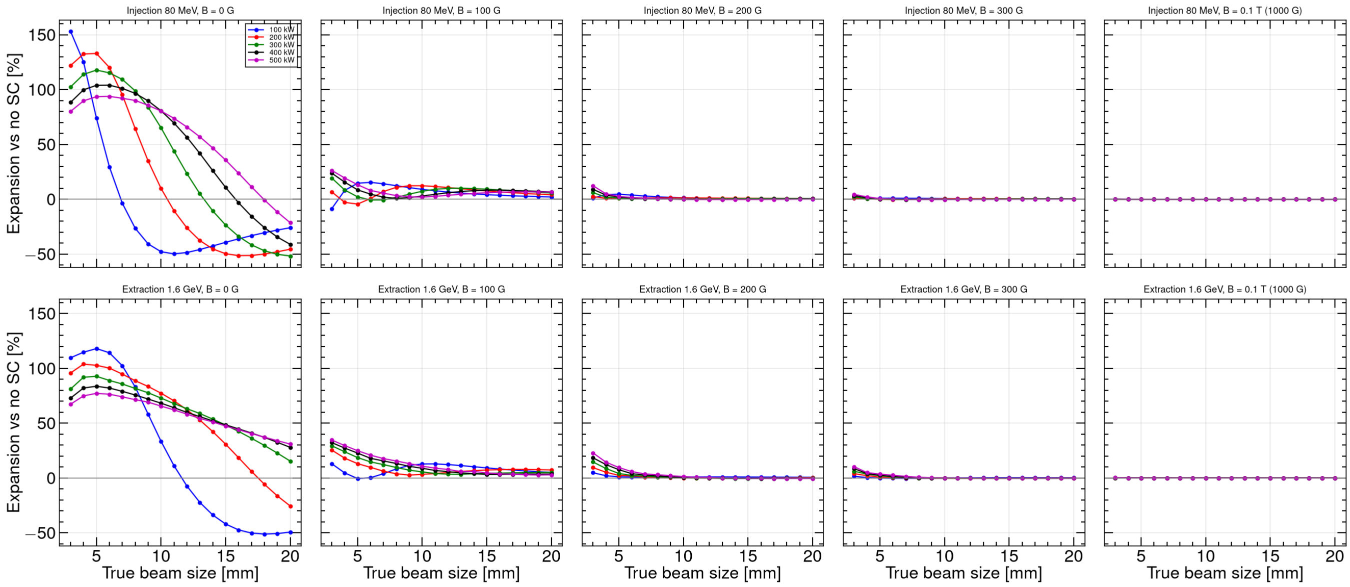

Figure 4. Expansion vs true beam size at the same B points.

At B=0 obtained σ is **not** a monotonic function of true σ (focusing vs over-kick). From **200 G** the points lie on the diagonal except the tiniest beams. At **300 G**, injection 3 mm is still +1% (100 kW) to +4% (500 kW). **0.1 T** puts 3, 10, and 20 mm on the diagonal at every power.

3.3 Block C — cage voltage, including the 1 kV / 5 G fine scan

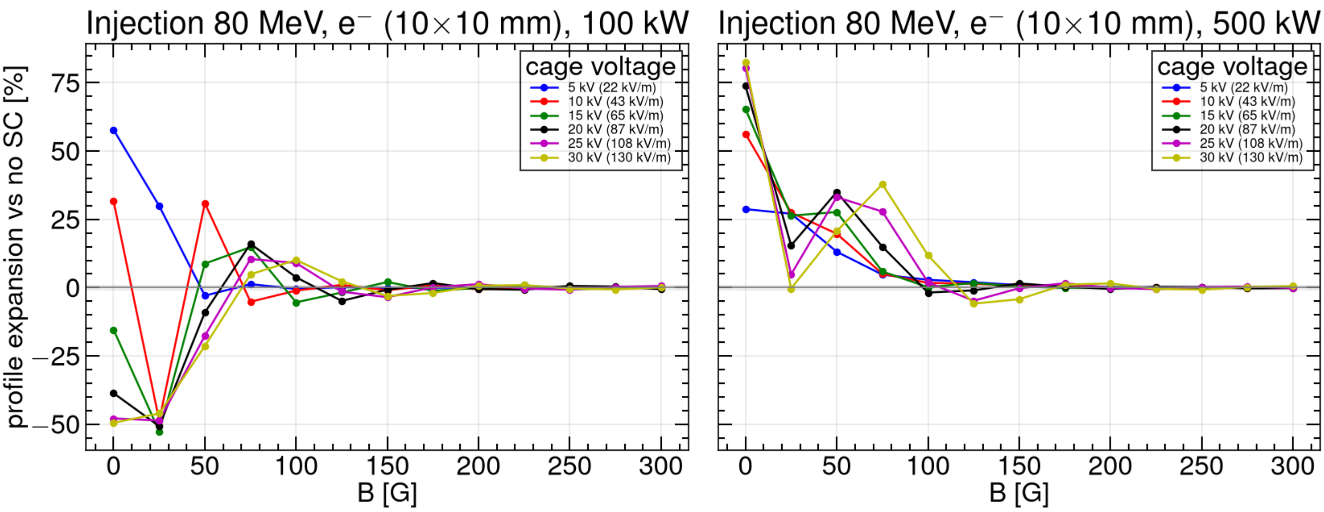

Figure 5. Coarse C: 5–30 kV / 5 kV on injection 10×10 mm.

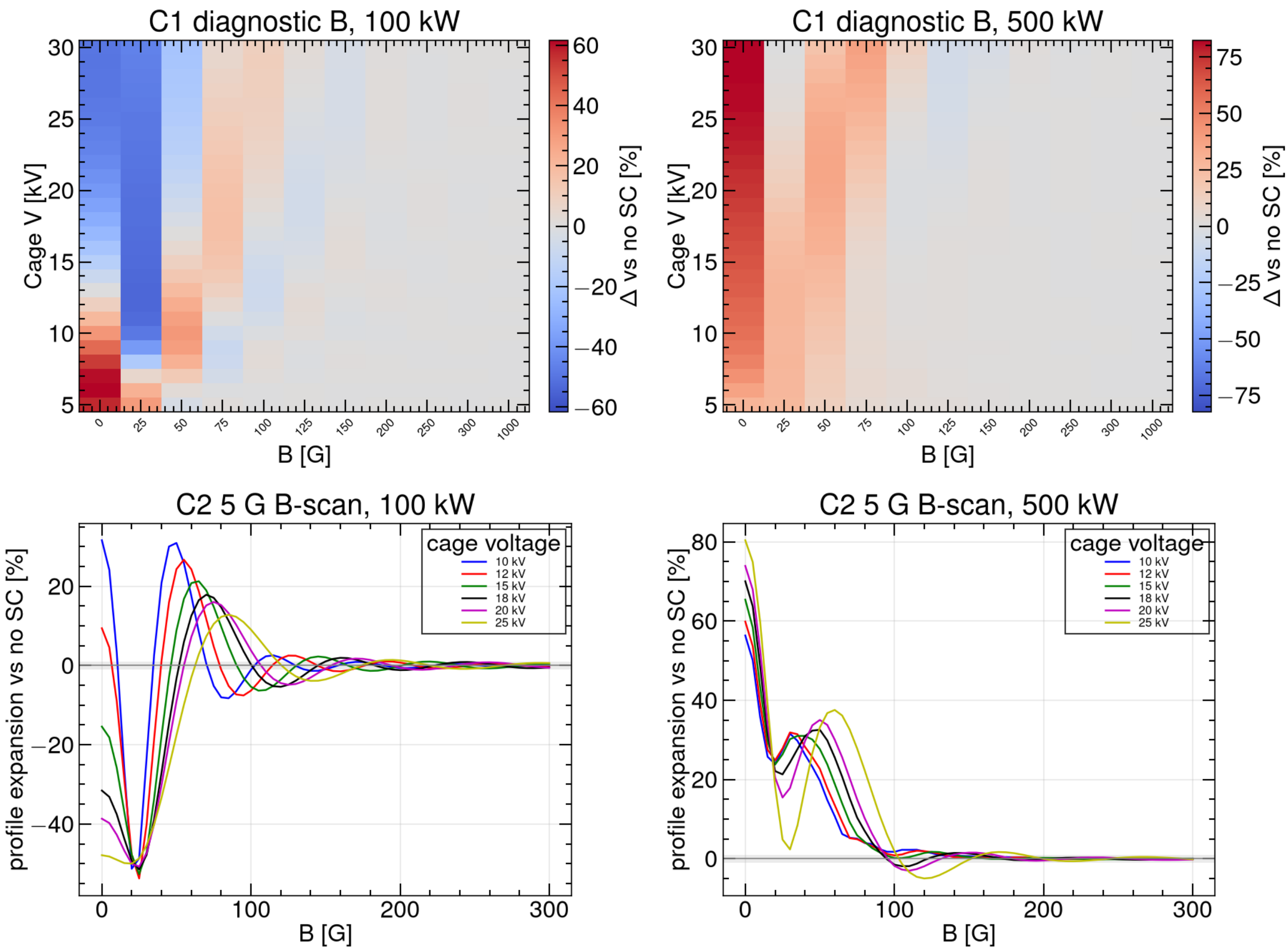

Figure 6. Fine C: 1 kV heatmap at diagnostic B, and 5 G B-scans at 10, 12, 15, 18, 20, 25 kV. At B=0, **100 kW crosses zero at 13 kV** (12 kV +9.4%, 13 kV ~0%, 14 kV −8.3%). **500 kW never flips**: Δ stays positive and grows with V (+29% at 5 kV to +83% at 30 kV).

The 5 G scans show the same cyclotron/ToF oscillation as Block A, with the first peak later in B as V rises. 1% threshold on 0–300 G:

V [kV]	100 kW	500 kW	Δ at 300 G (100 / 500 kW)
10	150 G	130 G	+0.15% / +0.02%
12	195 G	135 G	−0.31% / −0.05%
15	195 G	145 G	+0.47% / +0.03%
18	210 G	160 G	−0.47% / +0.01%
20	220 G	170 G	−0.57% / −0.19%
25 (Block A)	210 G	190 G	+0.61% / −0.15%

300 G still puts every scanned voltage ≲ 1%. Higher cage voltage needs more B at 100 kW; 500 kW recovers earlier at every V.

3.4 Block D — beam offset, including the 1 mm Δ y fine scan

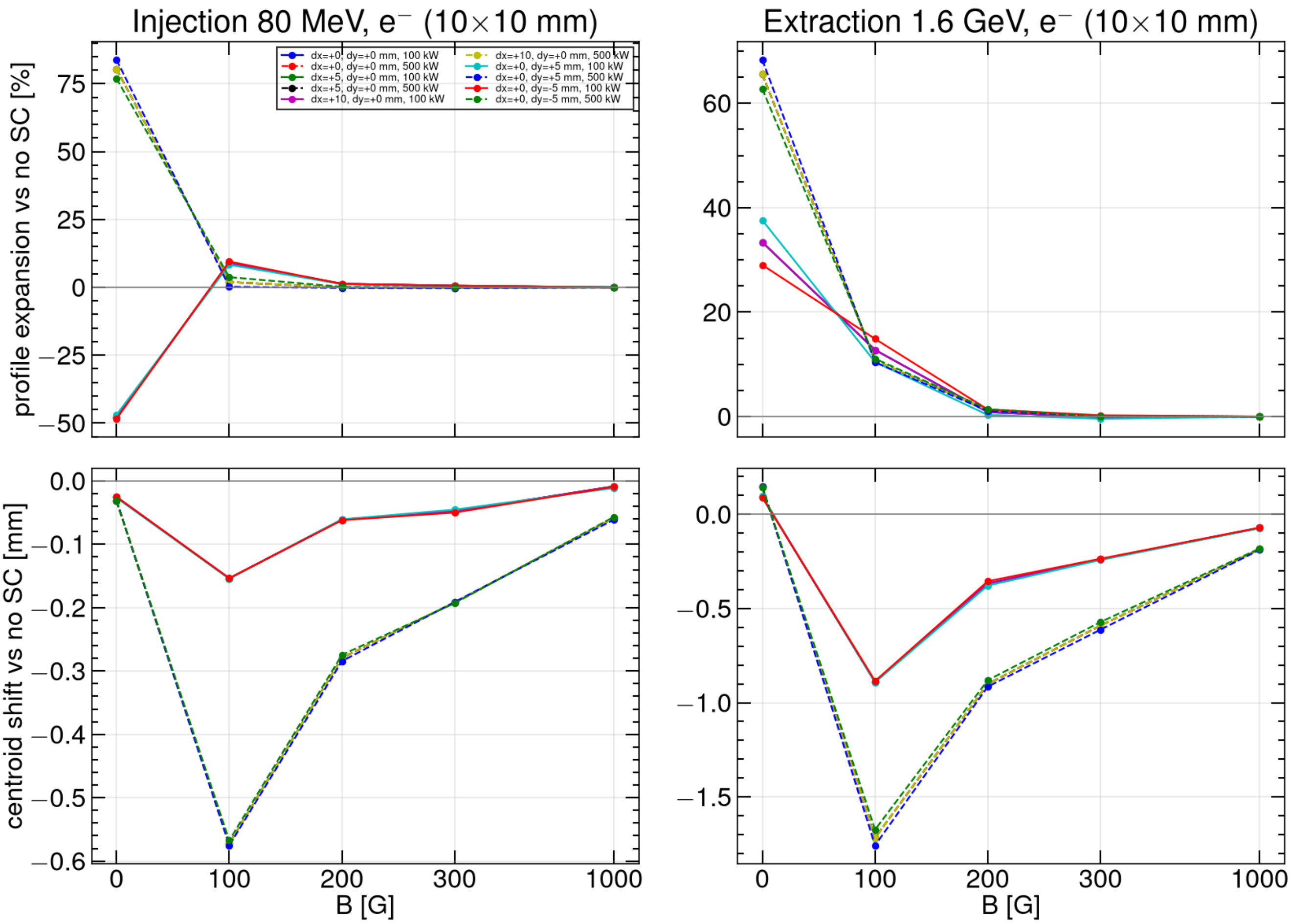

Figure 7. Coarse D: four offsets on 10×10 mm injection and extraction.

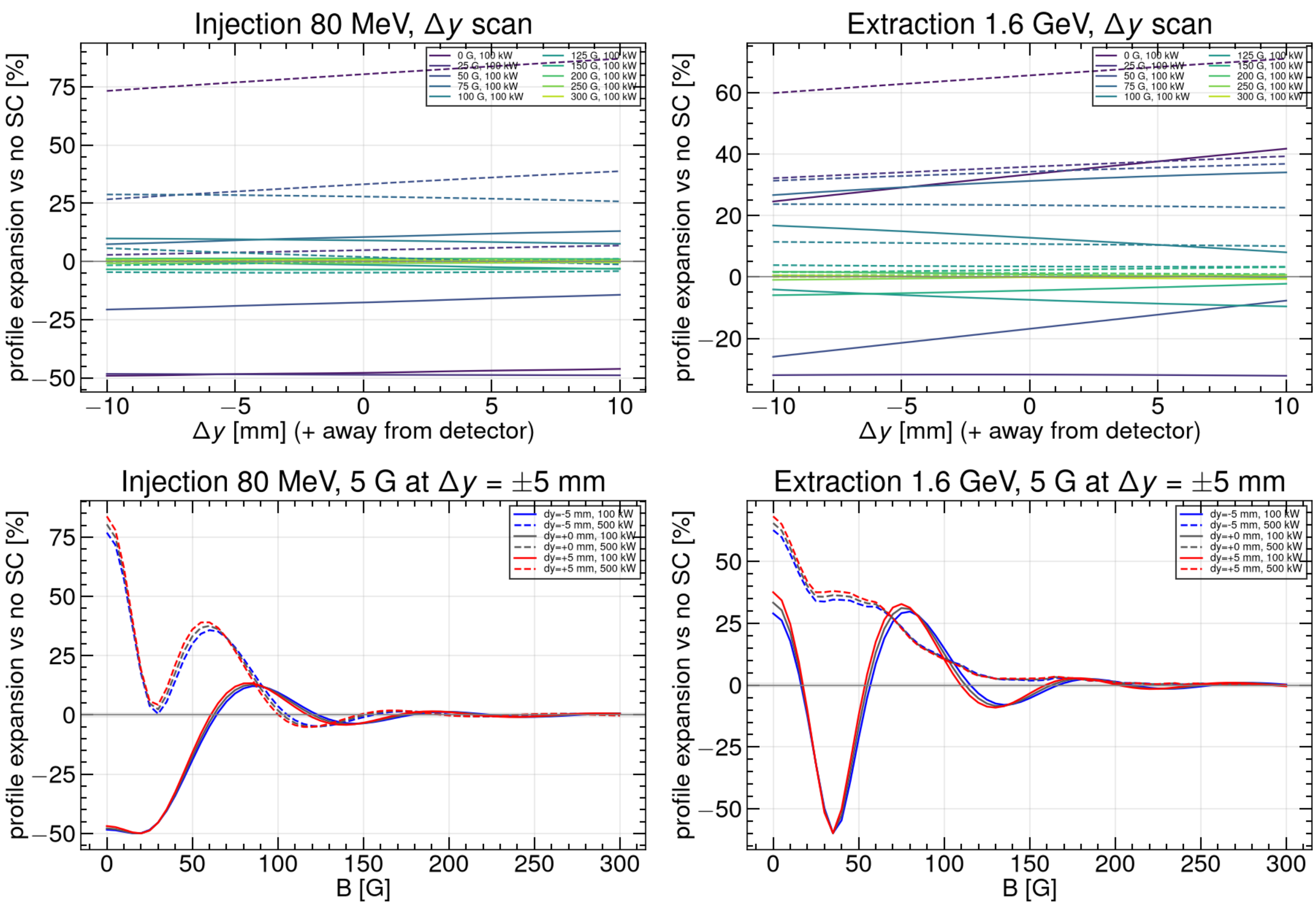

Figure 8. Fine D: expansion vs Δ y at diagnostic B, and 5 G B-scans at Δ y=±5 mm vs centred. Expansion vs Δ y is **smooth and nearly linear** from −10 to +10 mm. Toward the detector (Δ y<0) vs away (Δ y>0) changes Δ by a few percent at B=0 and by ≲ 1% once B=300 G. The ±5 mm 5 G scans overlay the centred Block A curves; the 1% threshold moves by at most ~10 G.

Family	Δ y=-5 mm	centred	Δ y=+5 mm
Inj. 100 kW	210 G	210 G	205 G
Inj. 500 kW	190 G	190 G	185 G
Ext. 100 kW	245 G	240 G	235 G
Ext. 500 kW	205 G	205 G	205 G

At **300 G** and **0.1 T**, expansion stays ≲ 1% for every offset. Centroid shift vs no-SC is ≲ 0.2 mm (injection) and ≲ 0.6 mm (extraction, 500 kW, 300 G). A few-mm orbit offset does not undo the B recovery. No Δ x scan: the cage is translation-invariant in x.

4. Ion mode

Status: complete. 2070 / 2070 runs. Round beams; H_2^+ / H_2O^+ / N_2^+. **No dense B-scan** — only 0, 200, 1000 G — because closed ion scans were already flat vs B.

Block	What is scanned	Beams	σ	B [G]	Power [kW]
A	Reference beams	inj. 25 mm; ext. 10 mm; inj. 10 mm	round	0, 200, 1000	100–500
B	Beam size	inj.; ext.	3–20 mm / 1 mm	0, 200, 1000	100–500
C	Cage voltage	inj. 10 mm	round	0, 200, 1000	100, 500
D	Beam offset	inj.; ext. 10 mm	round	0, 200, 1000	100, 500

SC-off reference: H_2^+ @ 100 kW only (mass-independent without bunch fields, shared).

4.1 Block A — reference beams vs B

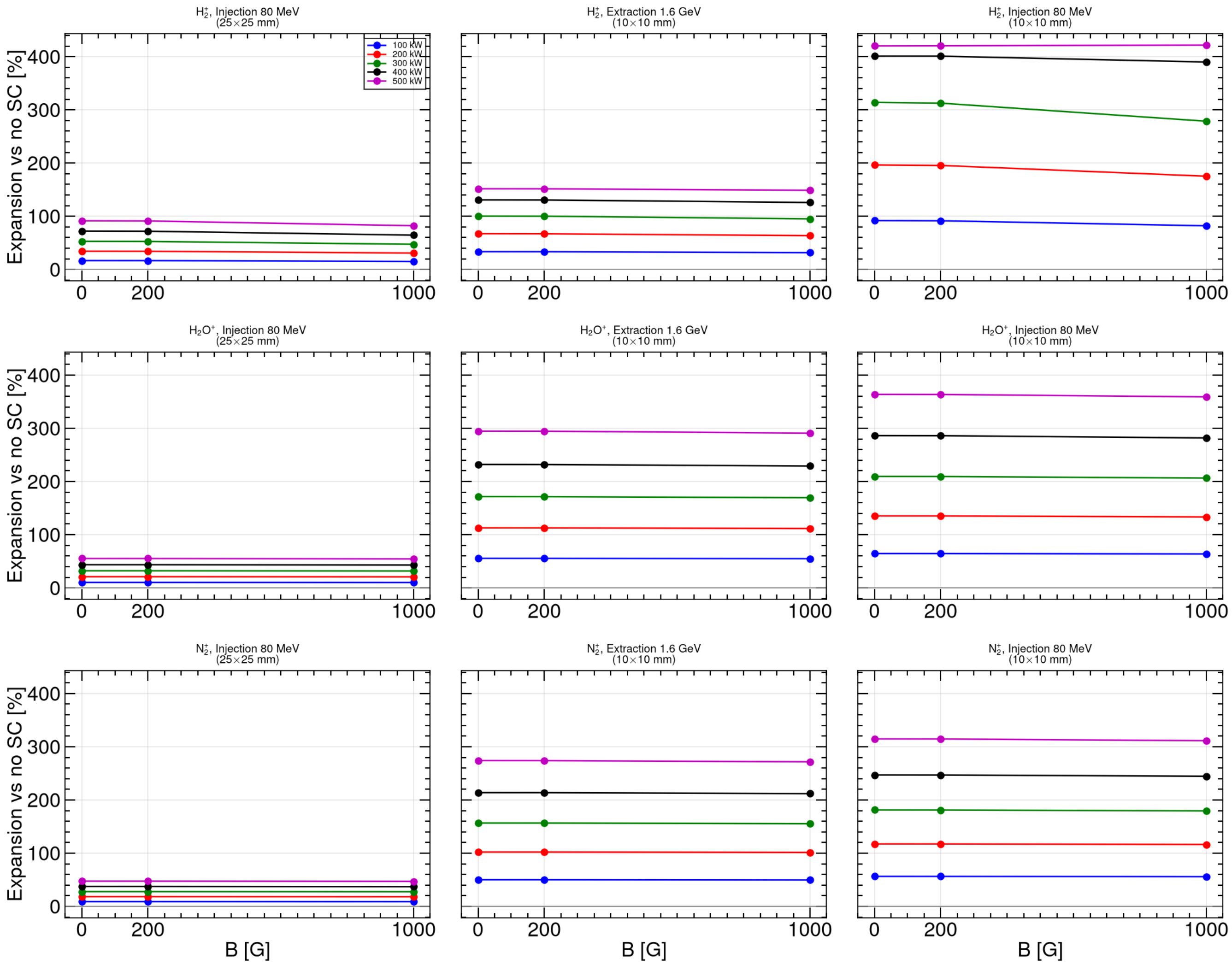

Figure 9. Expansion vs B for the three reference beams. Each row is a species; columns are painted injection (25×25 mm), extraction (10×10 mm), and small injection (10×10 mm).

Expansion is **essentially independent of B** between 0 and 200 G. The 0.1 T point is sometimes slightly lower but never restores the profile.

Expansion at 100 kW [%]:

Species	Inj. 25 mm 0 G	200 G	1000 G	Inj. 10 mm 0 G	200 G	1000 G	Ext. 10 mm 0 G	200 G	1000 G
H_2^+	+16.7	+16.7	+15.0	+92.0	+91.6	+82.1	+33.5	+33.4	+31.6
H_2O^+	+10.4	+10.4	+10.2	+64.7	+64.7	+63.8	+55.6	+55.6	+54.9
N_2^+	+8.9	+8.9	+8.9	+56.3	+56.2	+55.7	+49.9	+49.9	+49.5

Painted injection matches the elliptical-beam design-field numbers (~+16% H_2^+). The 10 mm injection family is the most distorted at every B.

Extraction H_2^+ (+33.5 % at 0 G) is a **lower bound**. Ions are generated 125 ns before the first field-carrying bunch, so H_2^+ has already drifted ≈ 40 mm when the kick arrives. A line-charge model with aligned generation gives ≈ +100 % (0 G, 100 kW). H_2O^+ / N_2^+ barely move in 125 ns and are close to the whole-bunch limit. Injection is correctly aligned (both centres at 480 ns). Details and the correction recipe: LITERATURE_REVIEW.md §§3.5–3.7.

4.2 Block B — size 3–20 mm

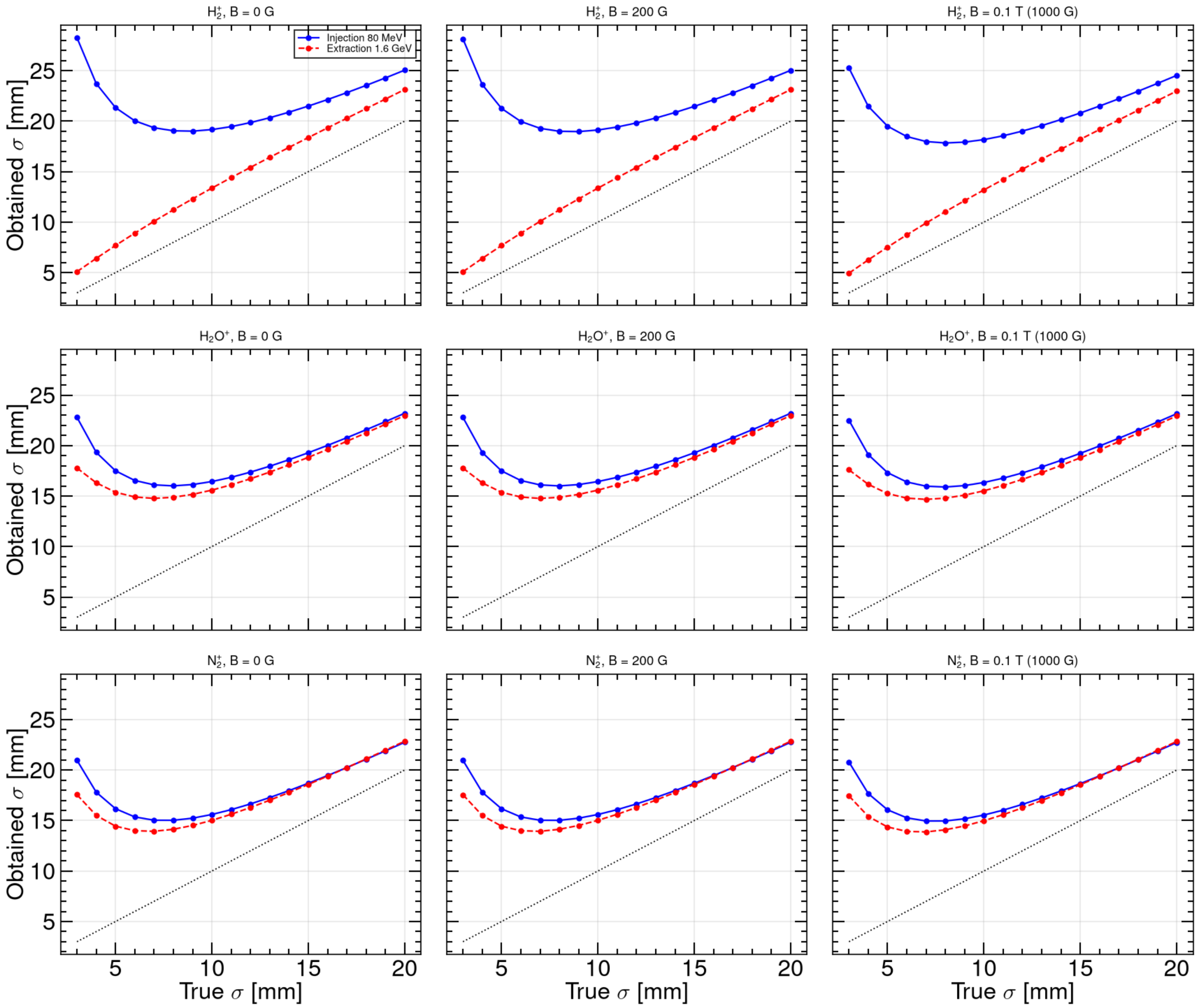

Figure 10. Obtained σ vs true σ at 0.1 T, 100 kW. Dashed line: obtained = true. Every point lies **above** the diagonal.

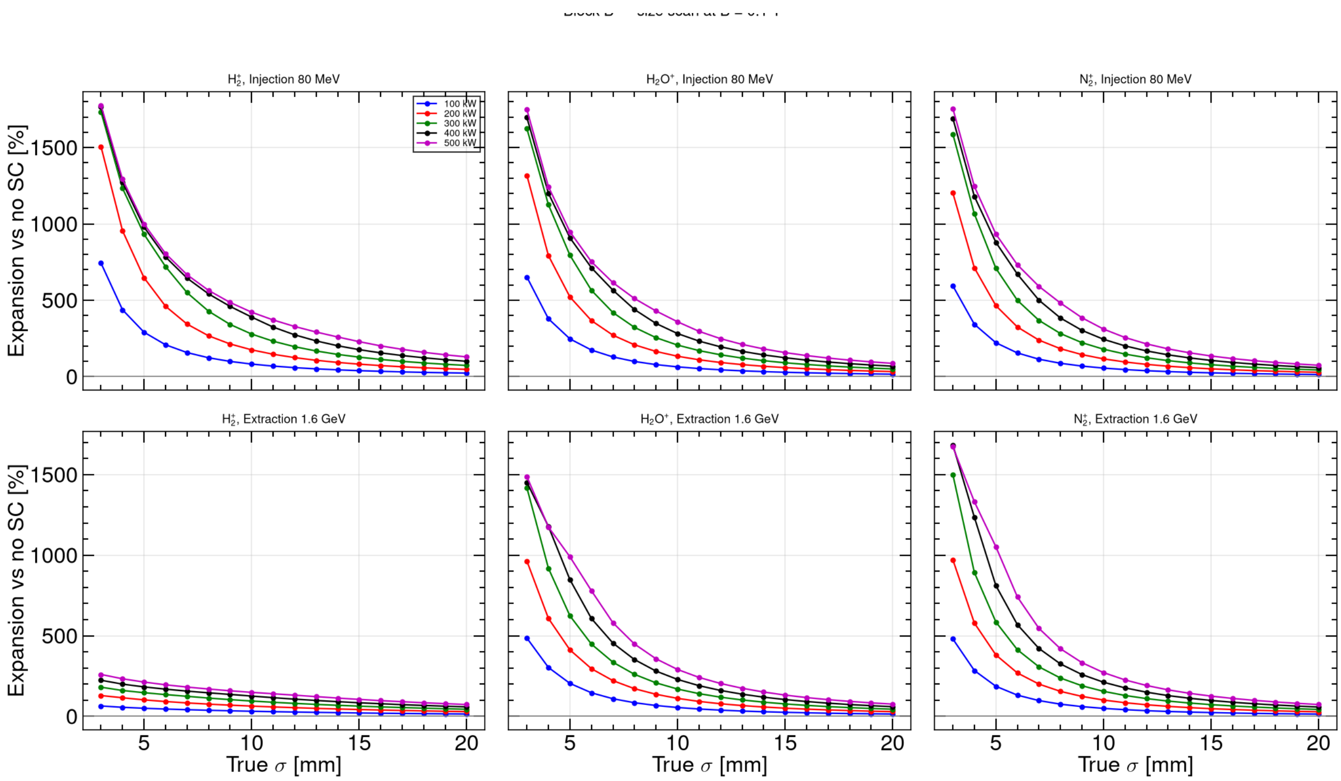

Figure 11. Expansion vs true σ at 0.1 T for 100–500 kW. Smaller beams and higher power inflate more.

Obtained size at σ_x = 10 mm, 0.1 T, 100 kW:

Species	Inj. obtained	Inj. expansion	Ext. obtained	Ext. expansion
H_2^+	18.2 mm	+82%	13.2 mm	+32%
H_2O^+	16.4 mm	+64%	15.5 mm	+55%
N_2^+	15.5 mm	+56%	15.0 mm	+50%

H_2^+ injection, σ=10 mm, 0.1 T vs power:

100 kW	200 kW	300 kW	400 kW	500 kW
+82% (18.2 mm)	+175% (27.5 mm)	+279% (37.8 mm)	+390% (48.9 mm)	+422% (52.1 mm)

Lighter ions expand more at injection ($\Delta v = q \in t \, E_{sc} \, dt/m$). At extraction the short 1.6 GeV bunch plus 0.1 T keeps H_2^+ nearer the diagonal; H_2O^+ and N_2^+ stay in the cage longer and can inflate more. Worst case: **injection, σ=3 mm, 500 kW, 0.1 T** — obtained σ ≈ 56 mm (+1700%+) for all three species (cage-wall saturation).

4.3 Block C — cage voltage 5–30 kV

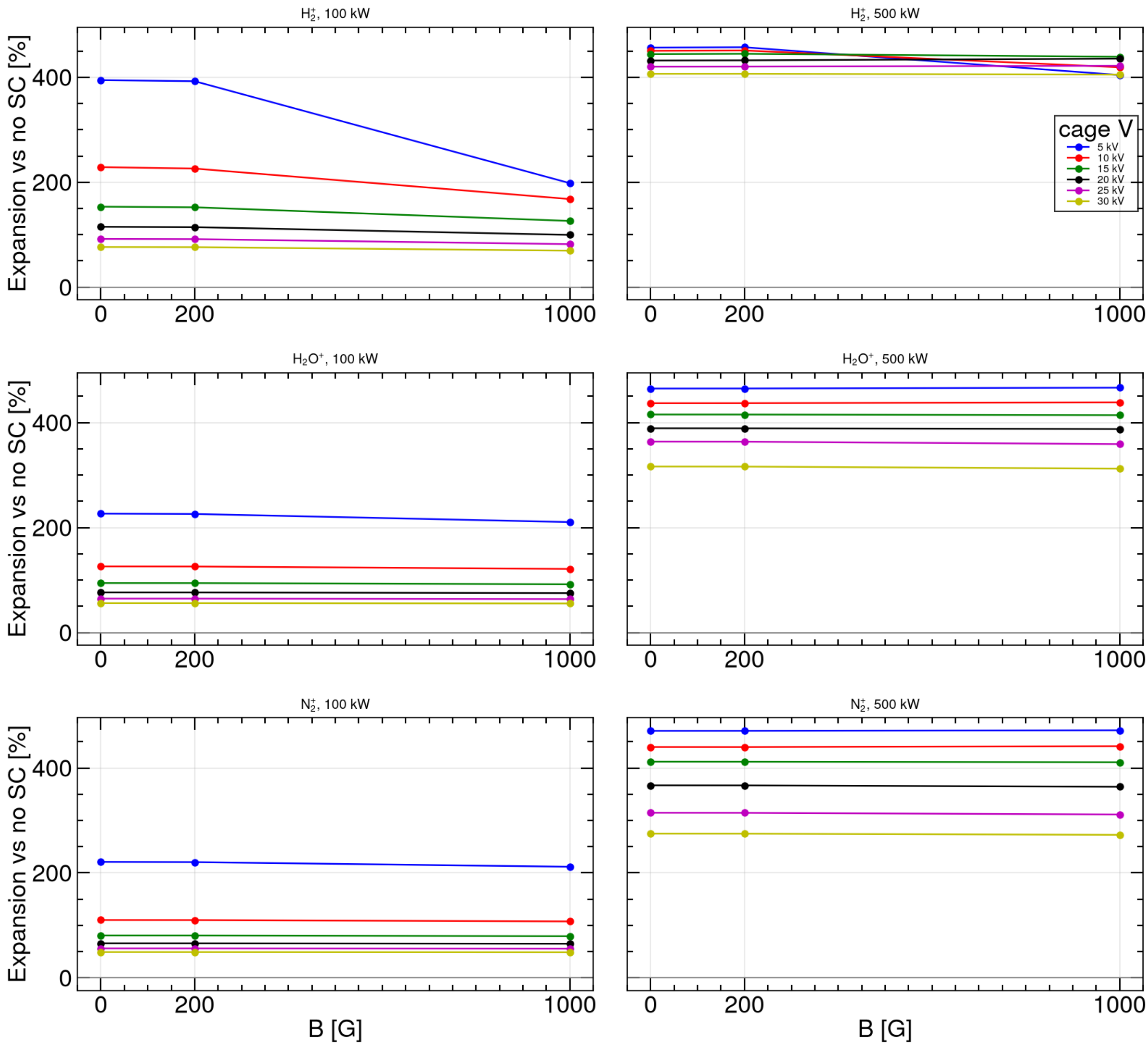

Figure 12. Expansion vs B for each cage voltage, injection 10×10 mm. Unlike e-mode, ion expansion **never changes sign**.

H_2^+ injection 10 mm, 100 kW, 0.1 T vs cage voltage:

5 kV	10 kV	15 kV	20 kV	25 kV	30 kV
+198%	+168%	+126%	+100%	+82%	+70%

Higher V shortens ion drift and reduces the integrated kick. At 500 kW the same trend holds at much larger expansion (+300% to +800% range).

4.4 Block D — beam offset

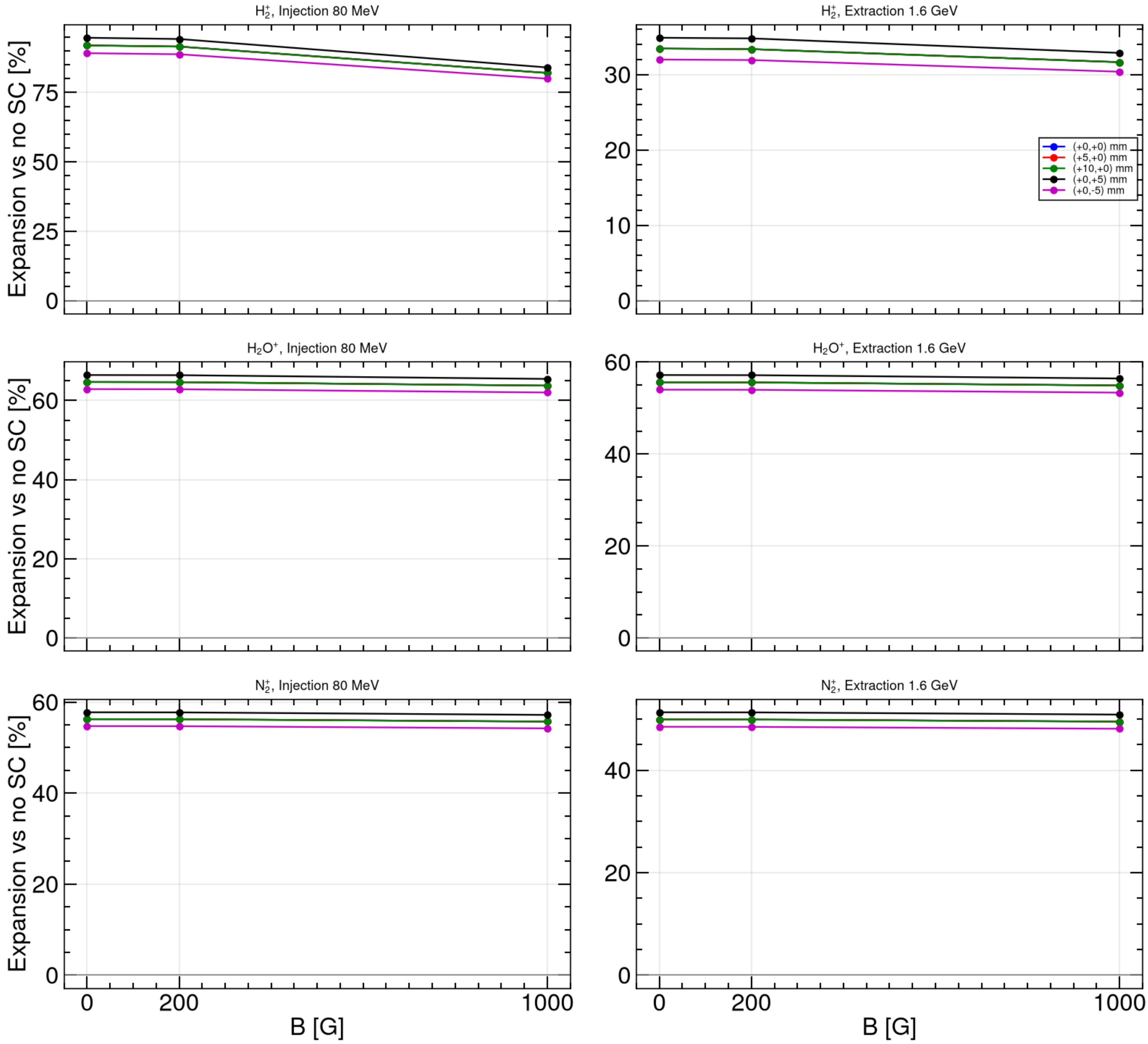

Figure 13. Expansion vs B for each offset (100 kW). Δ x=+5 and +10 mm overlay the centred beam. Δ y=±5 mm changes expansion by a few percent.

H_2^+ injection 10 mm, 0.1 T, 100 kW vs offset:

centred	Δ x=+5	Δ x=+10	Δ y=+5	Δ y=-5
+82.1%	+82.1%	+82.1%	+84.0%	+80.0%

Toward the detector (Δ y<0): slightly less expansion. Centroid shift vs no-SC is ≲ 0.6 mm at extraction, 500 kW.

5. Electron vs ion

	Electron mode	Ion mode
Secondaries	e− (Voitkiv H_2)	H_2^+ / H_2O^+ / N_2^+
B grid	dense (5 G / 25 G)	0, 200, 1000 G only
Effect of B	oscillatory recovery; 300 G fixes the profile	flat; no recovery
Cage voltage	sign flip at 13 kV (100 kW, B=0); 500 kW never flips	always positive; higher V reduces kick
Beam offset	Δ x invariant; Δ y few %	same
Worst case	low B, wrong V polarity	small σ, high P, injection

Design 0.1 T	conservative for electrons	still +15% to +80% (10–25 mm, 100 kW)

Operate the CSNS RCS IPM in electron mode with B≳ 300 G. Ion-mode profiles remain space-charge inflated at every scanned B, V, size, power, and offset; use them only with a correction model, or restrict ion ToF to species identification rather than size.

6. Data products

File	Contents
`output/csns_emode_bscan300_summary.csv`	E-mode Block A
`output/csns_emode_size_summary.csv`	E-mode Block B
`output/csns_emode_voltage_summary.csv`	E-mode Block C (coarse)
`output/csns_emode_offset_summary.csv`	E-mode Block D (coarse)
`output/csns_emode_voltage_fine_summary.csv`	Fine C
`output/csns_emode_offset_fine_summary.csv`	Fine D
`output/csns_imode_bscan_summary.csv`	Ion-mode Block A (135 rows)
`output/csns_imode_size_summary.csv`	Ion-mode Block B (1620 rows)
`output/csns_imode_voltage_summary.csv`	Ion-mode Block C (108 rows)
`output/csns_imode_offset_summary.csv`	Ion-mode Block D (180 rows)
`output/csns_imode_kick_model.csv`	Reduced line-charge model vs Virtual-IPM (0 G); aligned extraction column
`output/csns_review_scaling_checks.csv`	Literature scaling checks: Fine C zero spacing vs π m_e/(e ToF), Block B initial-velocity smear, ion size exponents, ion Δy vs drift length
`plots/csns_emode_*.png`	Figures 1–8
`plots/csns_imode_*.png`	Figures 9–13

Particle CSVs under output/emode/ and output/imode/ are gitignored. Lab notebook with the closed elliptical campaigns: REPORT.md. Ion-only write-up: IMODE_REPORT.md. Comparison with IPM papers 2006–2026: LITERATURE_REVIEW.md / LITERATURE_REVIEW.pdf.

Rebuild this PDF: python3 scripts/md_to_pdf.py --md CSNS_IPM_REPORT.md --pdf CSNS_IPM_REPORT.pdf --footer "CSNS RCS IPM — e-mode and ion-mode".